

# **Medical Terminology Identification Assignment**

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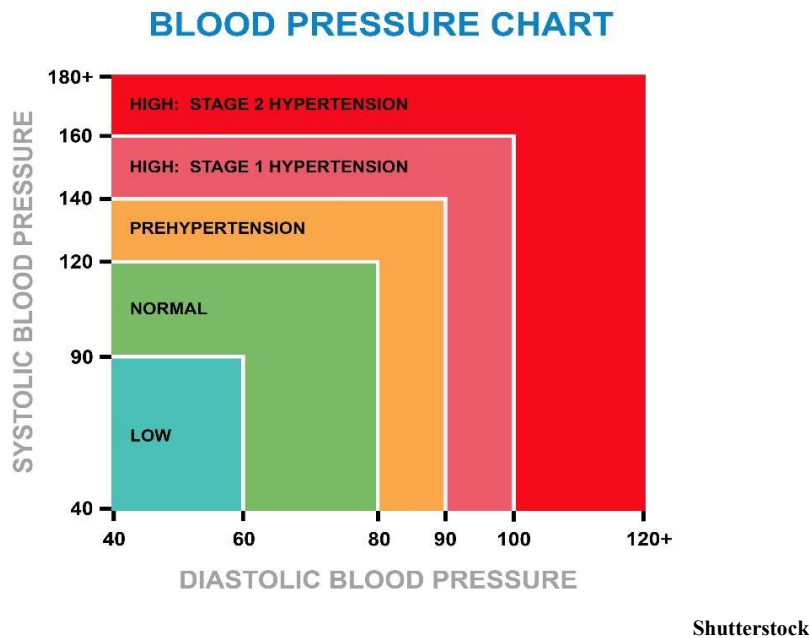
## Medical Terminology Identification

### Introduction

Medical terminology serves as the universal language of healthcare, allowing professionals to communicate complex physiological conditions and treatments with precision and efficiency. This assignment analyzes medical terminology from six recent publications in major medical journals. The sources cover key healthcare topics including diabetes care guidelines (American Diabetes Association, 2025), hypertension management (Flynn et al., 2025), acute coronary syndromes (Gulati et al., 2025), Long COVID (Davis et al., 2023), RSV vaccination (Melgar et al., 2024), and semaglutide therapy (Wilding et al., 2024). These articles represent current advances in chronic disease management, cardiovascular care, infectious disease prevention, and metabolic health, demonstrating how precise medical terminology enables effective communication among healthcare professionals.

### Term 1: Hypertension

- **Definition:** A chronic medical condition characterized by persistently elevated arterial blood pressure.
- **Context:** "The 2025 guidelines emphasize a team-based care approach... noting that lifestyle modifications remain the cornerstone of treatment for patients with stage 1 hypertension" (Flynn et al., 2025).
- **Importance:**



Hypertension is a primary modifiable risk factor for cardiovascular disease. Early identification is crucial to prevent end-organ damage, specifically to the kidneys, heart, and brain.

### Term 2: Myocardial Infarction

- **Definition:** Death of heart muscle tissue due to interruption of blood supply.
- **Context:** "New strategies to reduce bleeding risk in patients with acute coronary syndromes were incorporated into the updated management guidelines for myocardial infarction" (Gulati et al., 2025).
- **Importance:** This term is critical for diagnosing and treating heart attacks, a leading cause of mortality worldwide.

### Term 3: Diabetes Mellitus

- **Definition:** A metabolic disorder characterized by chronic hyperglycemia.

- **Context:** "For the first time, continuous glucose monitoring (CGM) is recommended for all adults with type 2 diabetes, including those not on insulin, to better manage glycemic levels and reduce complications" (American Diabetes Association [ADA], 2025).
- **Importance:** Understanding diabetes is essential for managing chronic disease and preventing organ damage.

#### Term 4: Dyspnea

- **Definition:** Difficult or labored breathing; often referred to as shortness of breath.
- **Context:** "Patients frequently reported persistent dyspnea upon exertion, even in cases where initial respiratory symptoms were mild." (Davis et al., 2023).
- **Importance:** Dyspnea is a primary symptom for cardiac and respiratory distress. Accurate documentation of dyspnea helps clinicians prioritize care and determine the severity of a patient's condition, particularly in emergency and pulmonology settings.

#### Term 5: Tachycardia

- **Definition:** A heart rate that exceeds the normal resting rate (usually over 100 beats per minute in adults).
- **Context:** "A subset of patients developed postural orthostatic tachycardia syndrome (POTS) as a complication of the viral infection." (Davis et al., 2023).
- **Importance:** Recognizing tachycardia is critical for identifying cardiovascular instability, arrhythmias, or shock. It acts as an early warning sign for various systemic issues, ranging from dehydration to heart failure.

#### Term 6: Agonist

- **Definition:** A substance that initiates a physiological response when binding to a receptor.
- **Context:** "Semaglutide functions as a Glucagon-like peptide-1 (GLP-1) receptor agonist to regulate appetite and insulin secretion." (Wilding et al., 2024).
- **Importance:** Understanding pharmacology terms like "agonist" (versus "antagonist") is fundamental for prescribing and understanding how medications interact with the body's cellular receptors to treat disease.

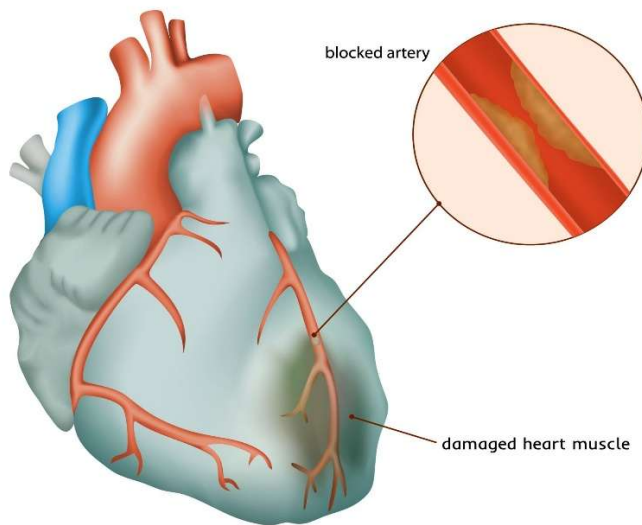
### Term 7: Subcutaneous

- **Definition:** Situated or applied under the skin.
- **Context:** "The medication was administered via a weekly subcutaneous injection." (Wilding et al., 2024).
- **Importance:** This term specifies a route of administration. Distinction between subcutaneous, intravenous, and intramuscular routes is crucial for nursing and pharmacy practice to ensure medication safety and efficacy.

### Term 8: Myocardial Infarction

- **Definition:** Irreversible death (necrosis) of heart muscle secondary to prolonged lack of oxygen supply (ischemia).
- **Context:** "The guidelines address outcomes in patients with delayed emergency care for myocardial infarction" (Gulati et al., 2025).
- **Importance:**

## Myocardial infarction



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Commonly known as a heart attack, recognizing this term and its signs is perhaps the most critical skill in acute medicine, as "time is muscle" regarding treatment outcomes.

### Term 9: Prophylaxis

- **Definition:** Action taken to prevent disease, especially by specified means or against a specified disease.
- **Context:** "The monoclonal antibody was approved for prophylaxis against respiratory syncytial virus in infants." (Melgar et al., 2024).
- **Importance:** Prophylaxis represents the cornerstone of preventive medicine.

Distinguishing between prophylactic (preventative) and therapeutic (curative) measures is essential for public health strategies and infection control.

### Term 10: Efficacy

- **Definition:** The ability to produce a desired or intended result, often used to describe the performance of a drug or vaccine in controlled clinical trials.
- **Context:** "The vaccine demonstrated 82% efficacy in preventing severe lower respiratory tract disease." (Melgar et al., 2024).
- **Importance:** Efficacy data drives evidence-based medicine. Healthcare providers must understand this metric to evaluate whether treatment is worth the potential risks and costs for their patients.

## Conclusion

This review demonstrates the importance of medical terminology in understanding healthcare literature and clinical practice. Each term plays a critical role in diagnosis, treatment, and healthcare innovation. Additionally, the growing integration of AI into healthcare highlights the need for precise terminology to ensure accurate interpretation and ethical application. Mastery of these terms enhances communication, supports evidence-based practice, and improves patient outcomes.

## References

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